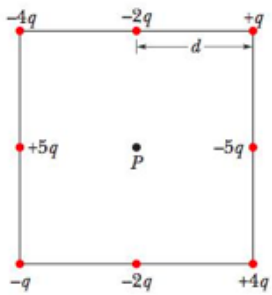
Discussion Questions

1



The diagram shows eight particles that form a square, with distance d between adjacent particles. What is the net electric potential at point P at the centre of the square (take the electric potential to be zero at infinity)?

Halliday, David. *Fundamentals of Physics Extended, 10th Edition*. John Wiley & Sons

2

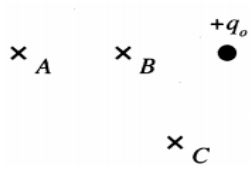
A positive point charge of magnitude q_0 is shown in the figure.

(a) Points B and C are located at a distance r from the charge. Point A is at a distance $2r$.

(i) Indicate the direction of the electric field at points A, B and C in the figure.

(ii) Compared to the absolute value of the work done in moving a small test charge from A to B: would the absolute value of this work done in moving the same test charge from B to C be *larger*, *smaller*, or *the same*?

(iii) Would the absolute value of the work done in moving the same test charge from A to C be *larger*, *smaller*, or *the same*?



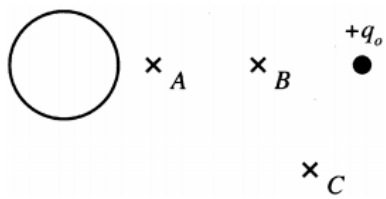
(b) A large metal sphere with zero net charge is now placed to the left of point A, as shown.

(i) Sketch the charge distribution on the metal sphere in the figure.

(ii) Has the magnitude of the electric field at points B and C *increased*, *decreased*, or *remained the same*? Explain.

(iii) Has the direction of the electric field at the points B and C changed? Explain.

(iv) Has the value of the electric potential V_A and at V_B *increased*, *decreased*, or *remained the same*?



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3

A solid conducting sphere of radius 1.0 m is given a charge of $+(4/9)\pi$ coulombs throughout its volume and allowed to reach equilibrium.

(a) What is the field and potential at its centre?

(b) Would your answer change if the sphere was an insulator? In what way(s)?

4

A parallel-plate capacitor is connected to a battery of electric potential difference V. If the plate separation is decreased, do the following quantities increase, decrease, or remain the same?

(a) the capacitor's capacitance,

(b) the voltage across the capacitor,

(c) the charge on the capacitor,

(d) the magnitude of the electric field between the plates.

5

Repeat the previous question for the case when the capacitor is first charged, and then disconnected from the battery. Do your answers change?

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6

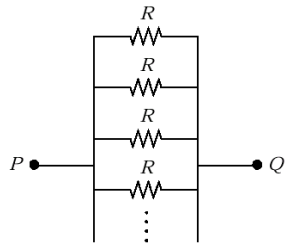
A capacitor is charged and left connected to the battery. A dielectric is then inserted, filling the capacitor. What happens to the quantities below when the dielectric is inserted?

- the electric field in the capacitor
- the voltage across the plates
- the charge on the plates
- the energy stored in the capacitor

7

The same capacitor is charged and disconnected from the battery. The same dielectric is again inserted, filling the capacitor. What happens to the quantities listed in Question 1 when the dielectric is inserted?

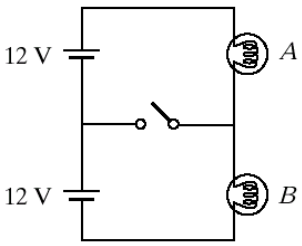
8



As more identical resistors R are added to the parallel circuit shown, does the total resistance between P and Q increase, decrease or remain the same?

9

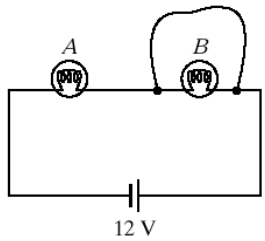
The light bulbs in the circuit are identical. When the switch is closed,



a) both go out.
b) the intensity of light bulb A increases.
c) the intensity of light bulb A decreases.
d) the intensity of light bulb B increases.
e) the intensity of light bulb B decreases.
f) some combination of 1–5 occurs.
g) nothing changes.

10

Two light bulbs A and B are connected in series to a constant voltage source. When a wire is connected across B as shown, bulb A:



a) burns more brightly.
b) burns as brightly.
c) burns more dimly.
d) goes out.

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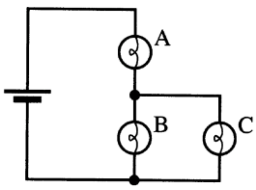
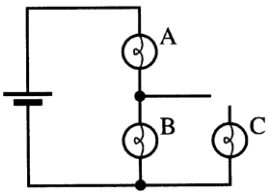
11



The globes in the circuit are identical; the battery is ideal.
ai) Rank bulbs A, B and C in order of brightness.
aii) Rank the voltages across the bulbs.
aiii) How do voltages across bulbs A and B relate to the battery voltage?
aiv) Is the voltage across bulb A *greater than, less than,* or *equal to* one half the battery voltage?



A student now cuts the wire between bulbs A and C, as shown.
bi) Rank bulbs A, B and C in order of brightness.
bii) Rank the voltages across the bulbs.
biii) How do voltages across bulbs A and B relate to the battery voltage?
biv) Is the voltage across bulb A *greater than, less than,* or *equal to* one half the battery voltage?

c) Analyse the following discussion between two students regarding the change in the circuit when the wire in the circuit is cut:

• Student 1*: "I think that bulb B will get brighter. Bulb B used to share the current with bulb C, but now it gets all the current. So bulb B will get brighter."*

• Student 2**:** *"I don't think so. Now there aren't as many paths for the current, so the resistance in the circuit has increased. Since the resistance in the circuit has increased, the current in the circuit decreases. Bulb B will get dimmer."*

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Problem-solving Questions

12

An electron in a TV picture tube is accelerated along the electric field (assumed uniform) between the filament and the screen, a distance of 50 cm. The field does 4.8×10^{-15} J of work on the electron. Assume the field is uniform.

a) Calculate the potential difference between filament and screen.
b) Calculate the electric field strength.
c) Calculate the force on the electron just after the filament? Just before the screen?

Halliday, David. *Fundamentals of Physics Extended, 10th Edition*. John Wiley & Sons

13

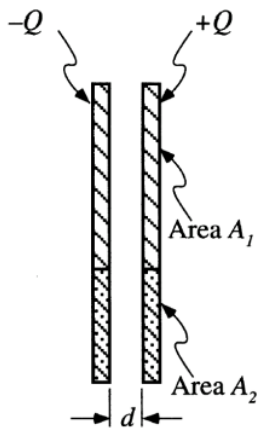
Water molecules have a permanent dipole moment of 6.2×10^{-30} C m. In a uniform electric field of 5×10^{6} V m^{-1}, calculate the:

- maximum torque that this field could cause, and
- the greatest change of electrical potential energy that could occur.

14

Find the potential at the centre of a copper ring with a charge density of 25 pC/m.

15 (mini-challenge question 1)



Two *very large* plates with area A_1 are held a distance d apart and have net charges -Q and +Q. Assume that all the charge is uniformly distributed on the inner surfaces of the plates.

a) Write an expression for the capacitance C in terms of A_1, d and ε_0.

Two initially *uncharged* plates of surface area A_2 are then attached to the original plates as shown to the right. Assuming d is very small.

b) Find the charge density on the plates and the voltage between them.

c) The right plate is now moved to the right, increasing the separation to D (still relatively small). Describe how each of the following quantities will change (if at all):

- the charge density on each plate
- the electric field both outside and between the plates
- the potential difference between the plates.

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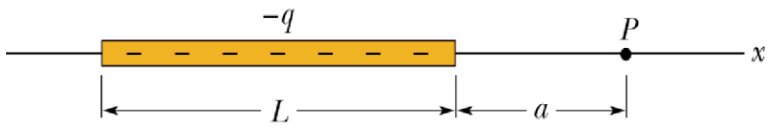
16 (mini-challenge question 2)

The diagram shows a *non-conducting* rod of length L with charge $-q$ uniformly distributed along its length. The linear charge density on the rod is given by λ and is equal to $-q/L$.



Calculate the electric potential at point P, a distance a from the end of the rod. (*NOTE* an integration is required.)

17

A parallel-plate capacitor is filled with a dielectric for which $\kappa = 5.5$. The area of each plate is 0.034 m^2, and the plates are separated by 2.0 mm. The capacitor will short out if the electric field between the plates is > 200 kN/C. In light of the energy density, what is the maximum energy that the capacitor can store?

Halliday, David. *Fundamentals of Physics Extended, 10th Edition*. John Wiley & Sons, 2013.

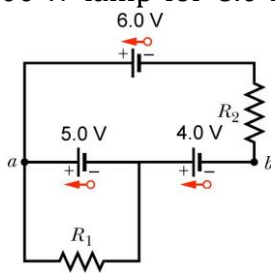
18

A standard AA battery can deliver about 2.0 W-hr of energy.

a) If a standard battery costs \$1.00, what is the cost of operating a 100 W lamp for 8.0 hr using such batteries?

b) Calculate the operating cost if energy is provided at 25 c per kW-hr

19



Find the current in each resistor and the potential difference between points a and b in the figure. Put R_1=100 Ω and R_2=50 Ω.

From *Fundamentals of Physics*, 6/e, Resnick, Halliday and Walker, © John Wiley 2001.

20

A battery with internal resistance (assumed constant) and unknown *emf* is connected to a 6.0 Ω resistor, and a current of 0.50 A flows through it. The resistor is then swapped for a 3.0 Ω resistor, and a current of 0.80 A flows. Calculate the internal resistance and *emf* of the battery.

21

A capacitor with plate area 2.0 m^2 and plate separation of 0.10 mm and an air dielectric (κ = 1) is charged from a 24 V battery. It is then filled with a castor oil dielectric (κ = 4.7). Calculate the difference in energy stored in the capacitor between the two arrangements.

Past examination questions

22

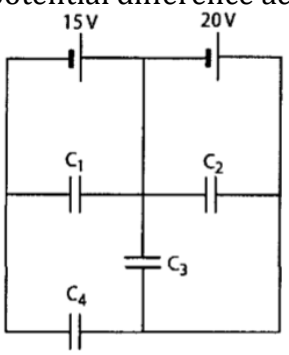
(a) A metal sphere of radius 20 cm has a net charge of -20 nC. There are no other charges present.

(i) What are the magnitude and direction of the electric field at the surface of the sphere?

(ii) Assuming that the electric potential is zero at infinity, what is the electric potential at the sphere's surface?

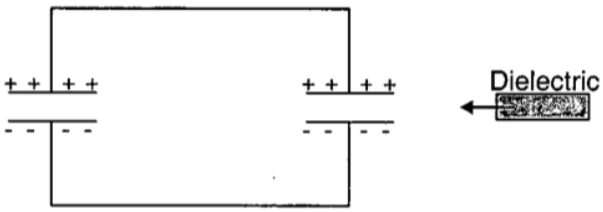
(iii) If an electron is released from a point at the surface of the sphere, and the force due to the sphere is the only force acting on it, how much kinetic energy does the electron gain as it is repelled from the surface?

(b) Determine the magnitude of the potential difference across each capacitor in the figure.



23

(a) Two identical capacitors with capacitance C, each carrying a charge Q, are connected as shown in the figure. A piece of dielectric material with a dielectric constant of κ is inserted and completely fills the gap between the plates of one of the capacitors.



Write down an expression for the new charge Q' on that capacitor.

24

A linear accelerator produces a beam of electrons that is pulsed. The current in each pulse is 0.5 A, and each pulse has a duration of 0.10 μs.

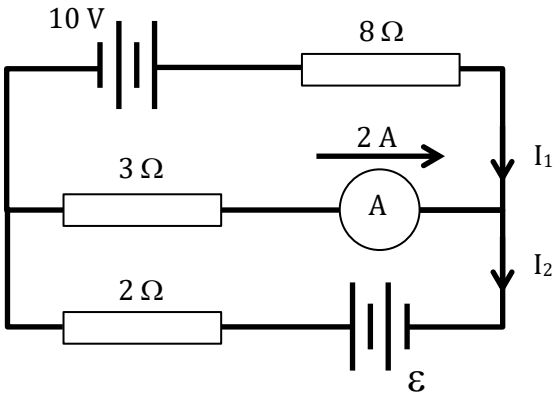
a) How many electrons are accelerated in each pulse?

b) What is the average current for an accelerator operating at 500 pulses per second?

c) If the electrons are accelerated to an energy of 50 MeV, what is the average power and peak power of the accelerator?

25

A circuit is set up with three resistors and two ideal batteries as in the diagram below. One of the batteries has an emf 10 V, but the emf of the second battery is unknown. An ammeter shows that 2 A is flowing through the 3 Ω resistor from left to right.

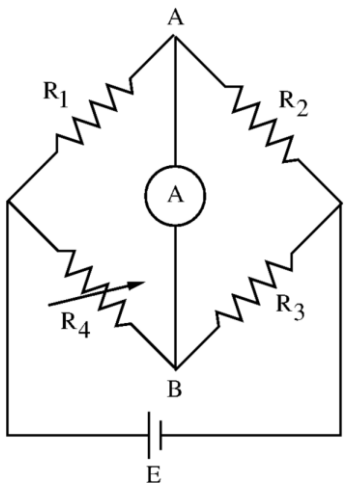


a) Using Kirchhoff's junction and loop laws, determine the emf, ε, of the second battery.

b) In the circuit above the 3 Ω resistor is replaced with a globe while a current of 2 A is still measured by the ammeter. The voltage across the globe is measured to still be 6 V and therefore the resistance of the globe is 3 Ω according to the equation $V = iR$. Can we assume the globe is ohmic? Explain your answer.

26

Four resistors are arranged as shown in the circuit diagram. R_1, R_2 and R_3 each have fixed resistance and R_4 is a variable resistor. A sensitive ammeter whose internal resistance is R_A is connected between points A and B in the circuit.



(a) By applying Kirchhoff's loop rule and junction rule, show that the ammeter will register no current flow when the variable resistor is adjusted so that its resistance satisfies

$$R_4 = R_1\left(\frac{R_3}{R_2}\right)$$

(b) If E = 5.0 V, R_1 = 3 Ω, R_2 = 2 Ω, R_3 = 1 Ω and R_4 = 1.5 Ω, what is the potential at A and B? (Assume the potential at the negative terminal of the cell is zero.)

Short solutions

Discussion Questions:

1 $-4kq/d = -q/\pi\varepsilon_0 d$ **2ai** field radially outward from charge **aii** smaller from B to C (zero)
2aiii the same **2bi** charge separation on metal sphere (induction) **bii** field at B & C both greater
2biii direction of field at B the same, changed at C **biv** potential has decreased at both points.
3a Conducting sphere: E at centre = 0 N/c, V at centre = V on surface (= $4\pi \times 10^9$ V)
3b Insulating sphere**:** E at centre still 0, but now potential will be greater than at the surface.
4a increase **b** the same **c** increase **d** increase
5a increase **b** decrease **c** the same **d** the same
6 ***E*** remains the same; V remains the same; Q increases; energy increases
7 ***E*** reduces; V reduces; remains the same; energy reduces
8 decrease **9** (g) **10** (a)
11ai &ii A > B = C **aiii** $V_A + V_B = V_{BATTERY}$ **aiv** greater than
bi & ii A = B; C is out **biii** $V_A + V_B = V_{BATTERY}$ **biv** equal to
ci & ii Student 1 is only partially correct, as is Student 2. The key is that the brightness depends on V_2/R, and as the resistance can (perhaps) be assumed to be the same, one should look at the voltage across the two lamps. The voltage across A decreases, across B increases.

Problem Solving Questions:

12a 30 kV **12b** 60 kV m^{-1} **12c** 9.6×10^{-15} N **13** 3.1×10^{-23} N m and 6.2×10^{-23} J **14** 1.41 V **15ab** $A_1\varepsilon_0/d$; $Q/(A_1 + A_2)$; $Qd/[\varepsilon_0 (A_1 + A_2)]$; **c** charge density unchanged; field outside zero and unchanged; field inside unchanged; V increases. **16** $V = -\frac{q}{4\pi\varepsilon_0 L}\log_e \frac{L+a}{a}$ **17** 6.6×10^{-5} J **18a** \$400 **18b** 20 c **19** Current in 50 Ω= 60 mA; current in 100 Ω = 50 mA; V_{AB} = 9.0 V **20** ε = 4.0 V and R_{int} = 2.0 Ω. **21** 1.89×10^{-4} J

Past Exam Questions:

22ai 4.5 kV m^{-1} radially inwards **aii** –900 V **aiii** 1.44×10^{-16} J;
b C_1: 15 V; C_2: 20 V; C_3: 20 V; C_4: 35 V.
23. $Q' = \frac{2\kappa Q}{\kappa+1}$ **24a** 3×10^{11} electrons **24b** 2.5×10^{-5}A **24c** Average power = 1250 W. Peak power = 25 MW. **25a** 14 V **25b** No – resistance may vary with voltage or current, need more data to confirm. **26a** *Hint: start by assuming no current through ammeter* **26b** $V_A = V_B = 2V$